

Computer Networks Lab01 Submission

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Task 01: Find the IP address of the computer you are using and the IP address version.

Command:

```
C:\Users\yashm>ipconfig
```

IP Address:

```
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

IP Address Version: IPv4

Task 02: Find the subnet mask of the computer you currently use, MAC address, the gateway, and whether DHCP is turned on.

Command:

```
C:\Users\yashm>ipconfig /all
```

Subnet Mask, Default Gateway, MAC Address:

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
Description . . . . . : Intel(R) Dual Band Wireless-AC 8265
Physical Address. . . . . : D4-3B-04-75-34-3D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::91a5:e657:820a:e2de%10(Preferred)
IPv4 Address. . . . . : 192.168.100.20(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, August 20, 2026 7:54:08 PM
Lease Expires . . . . . : Friday, August 21, 2026 7:54:07 PM
Default Gateway . . . . . : fe80::1%10
                             192.168.100.1
```

Task 03: Display the hostname of the computer and which protocol needs to be installed to access this command.

Command & Hostname:

```
C:\Users\yashm>hostname  
DESKTOP-A4HGUGP
```

Protocol: IP/TCP

Task 04: By which command do we check basic IP connectivity between two computers, check the connectivity with any other computer, and state the reason whether it's connected or not?

Command: In this i have used my own laptop's IP (Self Connectivity)

```
C:\Users\yashm>ping 192.168.100.20  
  
Pinging 192.168.100.20 with 32 bytes of data:  
Reply from 192.168.100.20: bytes=32 time<1ms TTL=128  
Reply from 192.168.100.20: bytes=32 time<1ms TTL=128  
Reply from 192.168.100.20: bytes=32 time<1ms TTL=128  
Reply from 192.168.100.20: bytes=32 time<1ms TTL=128  
  
Ping statistics for 192.168.100.20:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Reason:

Yes, it is connected because we received a reply from the destination computer.

Task 05: Find out which ports on your host are connected to applications. Connect the browser to some external web page before running the appropriate command.

Command:

```
C:\Users\yashm>netstat -ano
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	1464
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
TCP	0.0.0.0:1521	0.0.0.0:0	LISTENING	5980
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	9860
TCP	0.0.0.0:5432	0.0.0.0:0	LISTENING	7628
TCP	0.0.0.0:7680	0.0.0.0:0	LISTENING	1332
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	1156
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	792
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	2296
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING	3600
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING	4924
TCP	0.0.0.0:49675	0.0.0.0:0	LISTENING	1104
TCP	0.0.0.0:49680	0.0.0.0:0	LISTENING	12948
TCP	127.0.0.1:5500	0.0.0.0:0	LISTENING	5980
TCP	127.0.0.1:27017	0.0.0.0:0	LISTENING	7572
TCP	127.0.0.1:49682	0.0.0.0:0	LISTENING	12948
TCP	127.0.0.1:49683	0.0.0.0:0	LISTENING	12948
TCP	127.0.0.1:49684	0.0.0.0:0	LISTENING	12948
TCP	192.168.56.1:139	0.0.0.0:0	LISTENING	4
TCP	192.168.56.1:2030	0.0.0.0:0	LISTENING	5940
TCP	192.168.100.20:139	0.0.0.0:0	LISTENING	4
TCP	192.168.100.20:7680	192.168.100.47:58972	TIME_WAIT	0
TCP	192.168.100.20:7680	192.168.100.47:58973	TIME_WAIT	0
TCP	192.168.100.20:49236	192.178.173.139:443	ESTABLISHED	2808

```
C:\Users\yashm>tasklist /FI "PID eq 2808"
```

Image Name	PID	Session Name	Session#	Mem Usage
chrome.exe	2808	Console	2	48,692 K

Reason:

My laptop is using local port 49236 to connect to a remote web server through port 443 (HTTPS).

Task 06: Find the path of routers to at least three websites(of your own choice). What is its IP address? How many hops are involved in the path?

Command:

1.

```
C:\Users\yashm>tracert www.google.com

Tracing route to www.google.com [142.251.156.119]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  192.168.100.1
  2     3 ms     4 ms     2 ms  192.168.151.191
  3     *         *         *    Request timed out.
  4     4 ms     3 ms     3 ms  192.168.32.5
  5     3 ms     3 ms     3 ms  192.168.253.253
  6     4 ms     3 ms     4 ms  119.159.240.165
  7     4 ms     3 ms     3 ms  10.253.4.68
  8     6 ms     5 ms     3 ms  10.253.4.22
  9    22 ms    22 ms    22 ms  72.14.219.254
 10    21 ms    21 ms    21 ms  142.251.156.119

Trace complete.
```

2.

```
C:\Users\yashm>tracert www.github.com

Tracing route to github.com [20.207.73.82]
over a maximum of 30 hops:

  1     2 ms     1 ms     4 ms  192.168.100.1
  2     5 ms     2 ms     2 ms  192.168.151.191
  3     *         *         *    Request timed out.
  4     3 ms     2 ms     2 ms  192.168.32.5
  5     3 ms     2 ms     3 ms  192.168.253.253
  6     4 ms     3 ms     5 ms  119.159.240.165
  7     4 ms     3 ms     3 ms  10.253.4.68
  8     3 ms     3 ms     3 ms  10.253.4.26
  9     *         *        21 ms  ae61-0.ier01.dxb20.ntwk.msn.net [104.44.36.139]
 10    52 ms     *        52 ms  po25.rwa03.dxb21.ntwk.msn.net [51.10.44.0]
 11    51 ms    50 ms    51 ms  be1012.owr01.dxb21.ntwk.msn.net [51.10.9.83]
 12    52 ms     *        68 ms  be11.owr01.bom32.ntwk.msn.net [51.10.50.69]
 13    66 ms     *        54 ms  be8.ibr01.pnq21.ntwk.msn.net [104.44.18.26]
 14    72 ms     *         *    ae104-0.rwa03.pnq21.ntwk.msn.net [104.44.20.52]
 15     *         *         *    Request timed out.
 16     *         *         *    Request timed out.
 17     *         *         *    Request timed out.
 18     *         *         *    Request timed out.
 19     *         *         *    Request timed out.
 20     *         *         *    Request timed out.
 21    52 ms    50 ms    50 ms  20.207.73.82

Trace complete.
```

3.

```
C:\Users\yashm>tracert www.chatgpt.com

Tracing route to www.chatgpt.com [104.18.32.47]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  192.168.100.1
  2     2 ms     3 ms     2 ms  192.168.151.191
  3     *         *         *    Request timed out.
  4     4 ms     6 ms     3 ms  192.168.32.5
  5     4 ms     2 ms     2 ms  192.168.253.253
  6     6 ms     *         *    111.119.161.74
  7     6 ms     3 ms     2 ms  104.18.32.47

Trace complete.
```

Task 07: A ping to 8.8.4.4 works but a ping to the machine's name "Syrin machine" fails. What could be wrong? Write reason in concise form (1 to 2 sentences).

Reason:

Because 8.8.4.4 is an IP address whereas "Syrin machine" is a hostname

Task 08: If the order of the wires on both ends of the RJ45 connector is the same, which type of cable is used?

Answer: Straight-through cable

Task 09: Can you connect a Router Ethernet port to the PC NIC using a straight-through cable?(Yes or No)

Answer: Yes

Task 10: Can you connect a Switch to a PC or server using a crossover cable?(Yes or No)

Answer: No

Task 11: Can you connect a Hub to a PC or server using a straight-through cable?(Yes or No)

Answer: Yes

Task 12: Find all other hosts available on the network.

Command:

```
C:\Users\yashm>arp -a

Interface: 192.168.100.20 --- 0xa
    Internet Address      Physical Address      Type
    192.168.100.1         5c-09-79-7e-72-71    dynamic
    192.168.100.51        50-76-af-8c-cb-61    dynamic
    192.168.100.141       74-97-79-2a-4b-ad    dynamic
    192.168.100.150       cc-15-31-71-1c-22    dynamic
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.56.1 --- 0x15
    Internet Address      Physical Address      Type
    192.168.56.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
```

Task 13: Check any port on TCP and show the results

Command:

```
C:\Users\yashm>netstat -an

Active Connections

Proto Local Address      Foreign Address    State
TCP   0.0.0.0:135         0.0.0.0:0          LISTENING
TCP   0.0.0.0:445         0.0.0.0:0          LISTENING
TCP   0.0.0.0:1521        0.0.0.0:0          LISTENING
TCP   0.0.0.0:5040        0.0.0.0:0          LISTENING
TCP   0.0.0.0:5432        0.0.0.0:0          LISTENING
TCP   0.0.0.0:49664       0.0.0.0:0          LISTENING
TCP   0.0.0.0:49665       0.0.0.0:0          LISTENING
TCP   0.0.0.0:49666       0.0.0.0:0          LISTENING
TCP   0.0.0.0:49667       0.0.0.0:0          LISTENING
TCP   0.0.0.0:49668       0.0.0.0:0          LISTENING
TCP   0.0.0.0:49675       0.0.0.0:0          LISTENING
TCP   0.0.0.0:49680       0.0.0.0:0          LISTENING
TCP   127.0.0.1:5500      0.0.0.0:0          LISTENING
TCP   127.0.0.1:27017     0.0.0.0:0          LISTENING
TCP   127.0.0.1:49682     0.0.0.0:0          LISTENING
TCP   127.0.0.1:49683     0.0.0.0:0          LISTENING
TCP   127.0.0.1:49684     0.0.0.0:0          LISTENING
TCP   192.168.56.1:139    0.0.0.0:0          LISTENING
TCP   192.168.56.1:2030   0.0.0.0:0          LISTENING
TCP   192.168.100.20:139  0.0.0.0:0          LISTENING
TCP   192.168.100.20:49416 4.213.25.240:443    ESTABLISHED
TCP   192.168.100.20:49839 57.144.149.32:5222   ESTABLISHED
TCP   192.168.100.20:50291 118.103.239.142:443 CLOSE_WAIT
TCP   192.168.100.20:50709 52.110.1.149:443     ESTABLISHED
TCP   192.168.100.20:50714 52.110.1.185:443     ESTABLISHED
TCP   192.168.100.20:52591 172.64.148.235:443   ESTABLISHED
TCP   192.168.100.20:54010 172.64.155.209:443   ESTABLISHED
TCP   192.168.100.20:55657 52.107.7.27:443      TIME_WAIT
```

Task 14: Display the number of datagrams sent and received.

Commands:

```
C:\Users\yashm>netstat -s

IPv4 Statistics

Packets Received                = 1292840
Received Header Errors          = 0
Received Address Errors         = 2620
Datagrams Forwarded              = 0
Unknown Protocols Received      = 8
Received Packets Discarded       = 5241
Received Packets Delivered       = 1291186
Output Requests                  = 623257
Routing Discards                 = 0
Discarded Output Packets         = 1042
Output Packet No Route           = 88
Reassembly Required              = 0
Reassembly Successful            = 0
Reassembly Failures              = 0
Datagrams Successfully Fragmented = 0
Datagrams Failing Fragmentation  = 0
Fragments Created                = 0
```